Problems with Fixed LR Optimizer

⇒ GD

→ uses entire dataset for each update

→ $\theta_{t+1} = \theta_t - \eta \nabla_\theta J(\theta)$

⇒ SGD

→ uses single example for each update

→ faster but very noisy updates

⇒ Mini-batch GD

→ uses batch of egs.

→ provides a balance btw. GD & SGD.

⇒ Momentum-based

→ SGD with momentum, NAG

→ accumulate velocity in consistent direc^{ns}.

⇒ Limita^{ns} with Fixed LR

1. all params have fixed LR

→ but in actual scenario, some features occur frequently while some others are sparse.

2. Chosing 1 particular LR is difficult

→ if LR is too large, then divergence,
if LR is too small, then slow
convergence.

3. LR scheduling is manual.
→ requires expert knowledge, tune
LR based on the problem.

→ Solution → Adaptive LR Methods
→ automatic LR tuning
→ per-parameter adaptaⁿ
→ better convergence